Graph-Based Polypharmacy Risk Networks for Adverse Event Prediction in ICU Patients: A MIMIC-IV Analysis

Aman Chandra H, Kriti Ravi Pandit, and Vidya M J
School of Computer Science and Engineering
RV University, Bengaluru, India
amanch.btech23@rvu.edu.in

***Abstract*—Polypharmacy – the concurrent use of multiple medications – is common in intensive care and is a recognized driver of adverse events, yet most machine-learning studies on MIMIC-IV target mortality prediction rather than modeling drug-interaction risk directly. We address this gap by constructing a population-level drug co-administration graph from 9,974 adult ICU stays and comparing three patient-level feature representations – node2vec graph embeddings, hand-crafted graph-theoretic statistics, and a one-hot tabular baseline – for predicting three polypharmacy-relevant outcomes: acute kidney injury (AKI, KDIGO-based), a delirium proxy, and a bleeding proxy. Across four classifiers and stratified 5-fold cross-validation, we find a mixed but interpretable pattern: the tabular baseline wins when given unrestricted access to drug identity (AKI AUROC 0.763, bleeding AUROC 0.864), while graph embeddings outperform a leakage-corrected tabular baseline for delirium (AUROC 0.876 vs. 0.858, paired bootstrap Δ=+0.018, 95% CI [0.011, 0.025], $p < 0.0001$), replicating on an independent, zero-overlap second subsample (Δ=+0.015). We further report a label-leakage correction identified during development (delirium label circularity with tabular drug features, corrected AUROC $0.944 \rightarrow 0.858$) and perturbation-based, reliability-filtered drug-risk rankings. This comparison is evidence for when graph-based representations add value over tabular baselines in EHR-based polypharmacy modeling.**

***Index Terms*—polypharmacy, graph embeddings, node2vec, MIMIC-IV, electronic health records, acute kidney injury, delirium, adverse drug events**

I. Introduction

Adverse drug events are among the most common preventable harms in critical care, and polypharmacy – the concurrent administration of multiple medications – is their most consistent upstream driver. Nearly every patient in the intensive care unit (ICU) receives multiple concurrent medications, and the resulting combinations are mechanistically linked to nephrotoxicity, delirium, and bleeding complications. Despite this, machine-learning research on the MIMIC-IV critical care database has concentrated overwhelmingly on a narrow set of targets – in-hospital mortality, length of stay, and readmission – leaving polypharmacy-specific adverse event modeling comparatively underexplored [1], [2]. As a result, the *combinatorial* structure of drug co-administration – which drugs are given together, how often, and to whom – is rarely modeled explicitly; most prior EHR risk models instead treat medication exposure as an unordered feature bag, discarding the co-administration structure entirely.

This paper asks a narrower, more falsifiable question than "can graphs model polypharmacy risk": under what conditions does a graph-based representation of drug co-administration outperform a conventional tabular representation built from the same underlying information? Answering this requires holding everything else fixed – cohort, labels, classifiers, cross-validation protocol – and varying only the feature representation, which is the design we adopt here. We build a population-level co-administration graph over generic drug names, learn node2vec [3] embeddings for each drug, and compare three patient-level feature representations head-to-head – graph embedding, graph-theoretic statistics, and a tabular one-hot baseline – for three clinically distinct outcomes: acute kidney injury (AKI), a delirium proxy, and a bleeding proxy.

Our contribution is threefold: (1) a three-way feature-representation comparison evaluated under matched conditions across three clinically distinct outcomes, rather than a single-outcome demonstration; (2) a documented and corrected label-leakage circularity between the delirium outcome definition and tabular drug features – a failure mode consistent with broader findings on leakage-driven overoptimism in health-care machine learning [4], [5] – together with a paired significance test validating the corrected result; and (3) a confirmatory replication of the headline result on an independent, zero-overlap cohort subsample, directly addressing the generalizability concern that any single-subsample result invites.

II. Related Work

MIMIC-based clinical prediction. The MIMIC-IV critical care database [6], and MIMIC-III before it, have enabled a large body of ICU outcome-prediction research, but that research is heavily concentrated on a narrow set of targets. Harutyunyan et al. [1] formalized this as a standard benchmark suite of four tasks – in-hospital mortality, length-of-stay, decompensation, and phenotyping – and the great majority of subsequent MIMIC-based deep learning papers evaluate against some subset of these same targets. This concentration has persisted into the MIMIC-IV era: recent work revisiting clinical outcome prediction specifically for MIMIC-IV continues to organize its tasks around diagnosis, procedure, length-of-stay, and routing prediction [2]. Polypharmacy-specific adverse event modeling, by contrast, has received comparatively little direct attention despite polypharmacy's well-established mechanistic link to nephrotoxicity, delirium, and bleeding; this concentration on mortality/LOS-style targets is the specific gap this paper addresses.

Graph and network representation learning. The embedding method we use, node2vec [3], extends DeepWalk [7] by generalizing random-walk sampling with biased second-order walks, learning a vector representation for each node such that nodes appearing in similar random-walk contexts receive similar embeddings. This family of methods was first brought into the clinical setting by Med2Vec [8], which learns medical-code embeddings from the co-occurrence structure of codes within a clinical visit – structurally close to how we treat drug co-occurrence within an admission – and by GRAM [9], which supplements code embeddings with a medical ontology graph via attention to address data sparsity. More recent work has extended graph-based EHR representation to heterogeneous, multi-task settings: MulT-EHR mines relations across a heterogeneous graph for simultaneous multi-task prediction – drug recommendation, length-of-stay, mortality, and readmission – on MIMIC-III and MIMIC-IV [10]. All of this work targets diagnosis/procedure codes or general EHR heterogeneity broadly, rather than drug co-administration specifically, and none runs a matched embedding-vs-tabular-baseline comparison of the kind we report here.

Polypharmacy and drug-interaction graph modeling. The most directly related prior work is Decagon [11], which models polypharmacy side effects with a graph convolutional network over a multimodal graph of protein-protein interactions, drug-protein targets, and drug-drug interaction edges labeled by side-effect type. Decagon operates at the molecular/pharmacological level – predicting which side effect a drug *pair* may cause from biological interaction data – whereas our graph is built directly from *observed* patient-level co-administration in ICU stays and predicts *patient*-level clinical outcomes (AKI, delirium, bleeding) rather than drug-pair side-effect labels. A recent survey of machine-learning approaches to drug-drug interaction prediction documents the rapid growth of graph neural network methods in this space alongside persistent challenges around model interpretability, generalizability, and integration with clinical workflows [12] – but this literature is overwhelmingly oriented toward predicting drug-pair interaction *labels* from molecular or knowledge-graph structure, not toward predicting patient outcomes from observed co-administration. The two framings are complementary: Decagon's molecular graph could in principle be used to validate or explain associations our co-administration graph surfaces empirically, which we note as a concrete direction for future work rather than attempt here.

Outcome-specific prior work. Prior machine-learning work on our three outcomes is extensive but methodologically disconnected from the graph-representation question we ask.

TABLE I
COHORT ATTRITION.

Filtering step	n (admissions)
Adult patients, first ICU stay per admission	85,242
ICU LOS ¿= 24h	68,013
Random subsample (seed=42, compute/cost constraint)	10,000
¿= 3 distinct generic drugs administered	9,974
Final cohort (all labels attached)	9,974

A systematic review of AKI prediction on MIMIC found performance varying widely (AUROC 0.49–0.99) with a high risk of bias in most models [13]; ICU delirium has been predicted with screening tools using structured variables and CAM-ICU-based labels similar to our proxy [14]; and freedom from red-blood-cell transfusion has been used as a bleeding proxy for ICU patients in MIMIC [15], structurally the labeling strategy we adopt. None of this work, however, evaluates graph-based drug representations against tabular baselines under a matched protocol – the comparison this paper contributes.

Data leakage in clinical machine learning. The circularity we identify and correct for the delirium outcome is an instance of a broader, well-documented failure mode. Davis et al. [4] frame when model features constitute label leakage in clinical prediction, arguing that leakage risk depends on the cadence, perspective, and applicability of a prediction – directly relevant to our case, in which the delirium label is partly defined by drugs that also appear as tabular features. More broadly, a survey of leakage-driven reproducibility failures across seventeen fields found leakage inflating results in 294 papers, sometimes enough to overturn a central claim [5]. We report our correction with a before/after comparison and a paired significance test in that same spirit of transparent auditing.

Interpretability. We use SHAP [16] for the tabular baseline's tree-based models, and a perturbation (leave-one-drug-out) analysis for the graph-embedding models, since raw embedding dimensions are not individually interpretable in the way SHAP requires.

III. METHODS

A. Cohort construction

We extracted adult patients (age $\geq$18) with a first ICU stay of length $\geq$24 hours from MIMIC-IV via Google BigQuery, yielding 68,013 eligible admissions. For BigQuery cost and local compute tractability, we drew a fixed-seed random subsample of 10,000 admissions (seed=42, $\approx$14.7% of the eligible cohort), then excluded admissions with fewer than 3 distinct administered drugs (no meaningful polypharmacy signal), yielding a final analytic cohort of $n = 9{,}974$. Table I reports the full filtering flow.

This subsampling is a stated scope limitation, not a hidden one: to test whether headline findings generalize beyond the specific subsample drawn, we separately replicate our primary result on an independent, zero-overlap second 10,000-admission subsample (Section IV-C).

TABLE II
POPULATION CO-ADMINISTRATION GRAPH SUMMARY STATISTICS.

Statistic	Value
Nodes (distinct drugs)	2,091
Edges (co-administered pairs)	160,276
Density	0.0733
Avg. clustering coefficient	0.0011
Connected components	5
Mean degree	153.3
Median degree	49.0
Max degree	1573

B. Outcome labels

We define three binary outcomes, each an explicit, documented proxy rather than an adjudicated clinical diagnosis:

- **AKI**: KDIGO stage ≥ 1 [17], computed via MIMIC-IV's derived `kdigo_stages` table. Positive in 7,312/9,974 (73.3%) of the cohort.
- **Delirium (proxy)**: administration of an antipsychotic or benzodiazepine commonly used for agitation (haloperidol, quetiapine, olanzapine, risperidone, ziprasidone, lorazepam, midazolam, diazepam) **AND** a positive CAM-ICU [18] chartevent (2 of the 4 CAM-ICU criteria are directly captured in MIMIC-IV chartevents: altered mental status and inattention; the full 4-criterion clinical algorithm is not reconstructed). Positive in 2,679/9,974 (26.9%).
- **Bleeding (proxy)**: a new packed red blood cell (PRBC) transfusion order after ICU day 2. Positive in 876/9,974 (8.8%).

All three are stated limitations in Section V-A: they are defensible, literature-grounded proxies, not gold-standard adjudicated outcomes.

C. Drug co-administration graph

Prescription records were normalized to approximate generic names via a documented, deterministic cleaning pipeline (dose/form-suffix stripping plus a small brand-to-generic synonym dictionary). For each admission, two drugs are considered co-administered if given on an overlapping calendar day. The *population-level* co-administration graph aggregates this across the full cohort: nodes are distinct generic drug names, and edge weight is the number of distinct admissions in which the pair was co-administered. Table II reports summary statistics; Figure 1 visualizes the top-60 highest-degree drugs, and Figure 2 shows the full degree distribution.

D. Feature representations

We compare three patient-level feature representations, computed for every admission:

1) **Graph embedding**: node2vec [3] is trained on the weighted population graph (walk length 40, 20 walks per node, embedding dimension 64, window 10, default $p{=}1, q{=}1$ – not hyperparameter-tuned, a deliberate choice for a fair, fixed-hyperparameter comparison across all three feature sets, discussed further in Section V). Each patient's feature vector is the mean embedding of their administered drugs.
2) **Graph-theoretic**: sum and max pairwise edge weight among a patient's drugs, and the density of their drug-induced subgraph.
3) **Tabular baseline**: one-hot indicators for every administered drug, plus age, sex, and a comorbidity count (distinct ICD diagnosis codes).

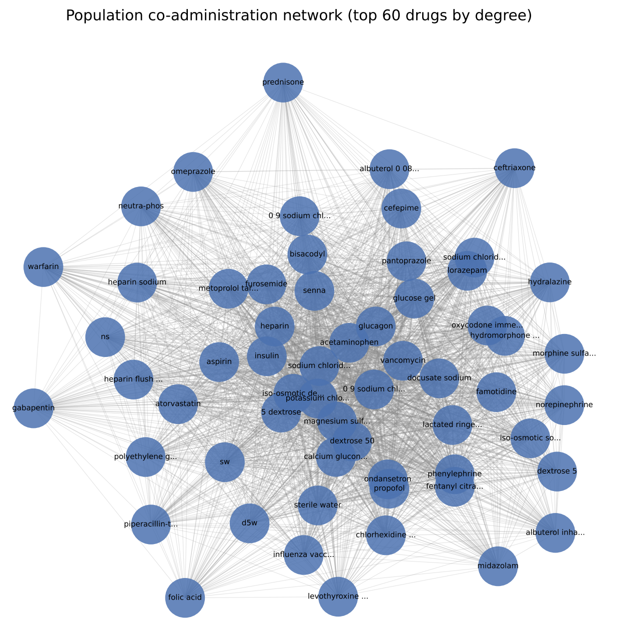

Fig. 1. Population co-administration network, top 60 drugs by degree. Node size scales with degree; edge width scales with co-administration weight.

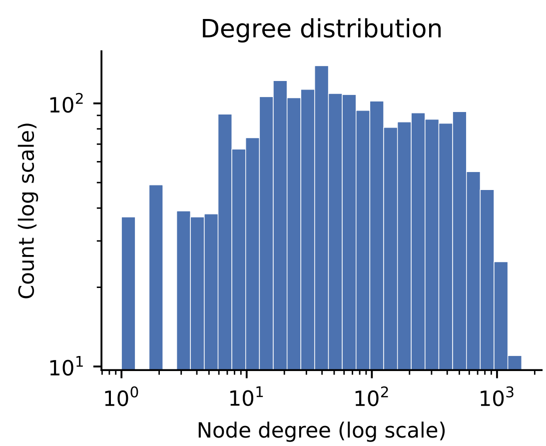

Fig. 2. Degree distribution of the population co-administration graph (log-log).

E. Modeling and evaluation

For every outcome $\times$ feature-set combination, we train four classifiers with fixed (untuned) hyperparameters – logistic re-

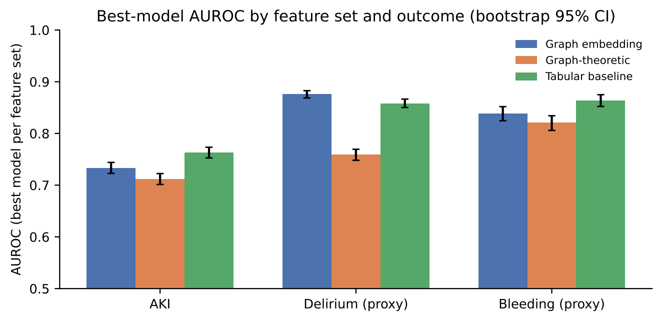

Fig. 3. Best-model AUROC by feature set and outcome, with bootstrap 95% CI error bars.

gression, random forest (300 trees), XGBoost [19] (300 trees, depth 4), and a 2-layer MLP (64, 32 units) – under stratified 5-fold cross-validation, implemented with scikit-learn [20]. We report AUROC, AUPRC, F1, and Brier score, with AUROC 95% confidence intervals from 1,000 bootstrap resamples of the pooled out-of-fold predictions.

F. Label-leakage correction

The delirium label is partly *defined* by administration of specific drugs (Section III-B); the tabular baseline's one-hot features include columns for those same drugs, so a tabular model can trivially reconstruct much of the label via this circularity rather than genuine predictive signal. We identified this after observing an implausibly high tabular AUROC (0.944, XGBoost) for delirium. We excluded the delirium-defining drug columns from the tabular feature set *for the delirium outcome only* (AKI and bleeding labels are not drug-defined and are unaffected) and refit all four tabular models, yielding a corrected AUROC of 0.858 (XGBoost).

IV. Results

A. Three-way feature comparison

Table III reports the full outcome $\times$ feature-set $\times$ model grid; Figure 3 summarizes the best-model AUROC per feature set and outcome. The comparison is mixed rather than a uniform win for any single representation: the tabular baseline achieves the highest AUROC for AKI (0.763, XGBoost) and bleeding (0.864, XGBoost), while the graph embedding achieves the highest AUROC for delirium (0.876, XGBoost) – exceeding even the leakage-corrected tabular result (0.858).

We caution that AUPRC for AKI (base rate 73.3%) should be interpreted relative to that high base rate rather than in isolation; AUROC is the more informative metric for this outcome, and we lead with it throughout.

B. Statistical significance of the delirium result

Because the embedding-vs-tabular delirium comparison is our headline claim, we tested it with a paired bootstrap on *identical* cross-validation fold assignments for both feature sets, rather than comparing two independent confidence intervals. Table IV reports a statistically significant advantage for the embedding (Δ=+0.0178, 95% CI [0.0112, 0.0249], $p < 0.0001$).

C. Confirmatory replication

To test whether this is specific to the subsample drawn, we repeated the entire pipeline – cohort draw, graph construction, node2vec training, and significance test – on a second, independent subsample with *zero overlap* with the original (seed=123, $n = 9{,}976$). The effect replicated (Table V): Δ=+0.0152, 95% CI [0.0081, 0.0218], overlapping the original CI.

D. Interpretability

For each outcome's best embedding model, we identify risk-associated drugs via perturbation analysis: for high-risk patients (top decile of predicted probability), we remove each drug from their exposure set, re-score, and average the resulting prediction delta across patients. To avoid reporting single-patient noise as signal, we exclude any drug supported by fewer than 10 high-risk patients before ranking (Figure 4). These rankings have *not* been cross-referenced against a clinical drug-drug-interaction database and should not be interpreted as validated interactions without that manual check.

V. Discussion

The three-way comparison does not support a blanket claim that graph embeddings outperform tabular baselines for polypharmacy risk modeling. Instead, it supports a more specific and, we think, more useful claim: a tabular representation with unrestricted access to raw drug identity has a strict informational advantage and wins when that access is available (AKI, bleeding), but once that advantage is removed by a genuine label-leakage correction (delirium), the compressed 64-dimensional graph embedding recovers signal that the reduced tabular baseline does not – and this advantage is both statistically significant on identical folds and replicates on an independent cohort subsample.

Our node2vec hyperparameters (p=1, q=1, no walk-length or dimension search) were fixed, not tuned, to keep the three-way comparison fair. This cuts in favor of the embedding result: an untuned graph embedding already outperforms a leakage-corrected tabular baseline, suggesting hyperparameter search would plausibly widen rather than narrow this gap – a direction for future work rather than a caveat on the present one.

A. Limitations

(1) All three outcome labels are proxies, not adjudicated diagnoses – most notably the delirium proxy captures 2 of 4 CAM-ICU criteria. (2) The analytic cohort is a fixed-seed random subsample of the eligible MIMIC-IV population, not the full 68,013-admission eligible cohort; while our delirium result replicates on an independent subsample, this was only tested for the headline claim, not for every outcome/feature-set combination. (3) Single-center, retrospective data (MIMIC-IV, derived from Beth Israel Deaconess Medical Center)

TABLE III
AUROC (95% bootstrap CI), AUPRC, F1, and Brier score for every outcome × feature-set × model combination. delirium_label / tabular rows are the leakage-corrected refit; pre-correction AUROC (before excluding delirium-defining drug columns) shown alongside for transparency (see Section III).

Outcome	Feature set	Model	AUROC	95% CI	AUPRC	F1	Brier
AKI	Graph embedding	random forest	0.733	[0.723, 0.744]	0.880	0.851	0.170
		xgboost	0.726	[0.716, 0.738]	0.878	0.845	0.172
		logistic regression	0.720	[0.709, 0.732]	0.869	0.846	0.172
		mlp	0.631	[0.618, 0.643]	0.806	0.778	0.296
AKI	Graph-theoretic	mlp	0.712	[0.701, 0.722]	0.866	0.842	0.175
		logistic regression	0.709	[0.698, 0.720]	0.863	0.843	0.176
		xgboost	0.706	[0.695, 0.716]	0.864	0.837	0.177
		random forest	0.662	[0.650, 0.673]	0.841	0.817	0.193
AKI	Tabular baseline	xgboost	0.763	[0.753, 0.773]	0.896	0.851	0.162
		random forest	0.759	[0.748, 0.770]	0.890	0.853	0.164
		logistic regression	0.686	[0.674, 0.697]	0.848	0.808	0.210
		mlp	0.659	[0.647, 0.670]	0.841	0.779	0.294
Delirium (proxy)	Graph embedding	xgboost	0.876	[0.869, 0.883]	0.735	0.625	0.121
		random forest	0.849	[0.841, 0.858]	0.710	0.569	0.133
		mlp	0.849	[0.842, 0.857]	0.664	0.621	0.182
		logistic regression	0.849	[0.841, 0.856]	0.659	0.587	0.135
Delirium (proxy)	Graph-theoretic	mlp	0.759	[0.748, 0.769]	0.561	0.433	0.161
		xgboost	0.755	[0.744, 0.765]	0.551	0.419	0.162
		logistic regression	0.738	[0.727, 0.749]	0.521	0.349	0.167
		random forest	0.718	[0.707, 0.729]	0.505	0.438	0.176
Delirium (proxy)	Tabular baseline	xgboost	0.858 (pre-corr. 0.944)	[0.850, 0.866]	0.704	0.606	0.128
		random forest	0.848 (pre-corr. 0.916)	[0.839, 0.856]	0.693	0.540	0.135
		logistic regression	0.784 (pre-corr. 0.855)	[0.773, 0.795]	0.557	0.565	0.173
		mlp	0.772 (pre-corr. 0.862)	[0.762, 0.782]	0.566	0.545	0.208
Bleeding (proxy)	Graph embedding	xgboost	0.838	[0.824, 0.852]	0.384	0.237	0.065
		logistic regression	0.810	[0.795, 0.825]	0.317	0.154	0.069
		random forest	0.787	[0.770, 0.803]	0.337	0.143	0.069
		mlp	0.783	[0.768, 0.798]	0.260	0.312	0.101
Bleeding (proxy)	Graph-theoretic	mlp	0.821	[0.806, 0.834]	0.320	0.109	0.069
		logistic regression	0.817	[0.802, 0.831]	0.318	0.153	0.070
		xgboost	0.807	[0.792, 0.821]	0.298	0.138	0.070
		random forest	0.761	[0.743, 0.776]	0.250	0.195	0.077
Bleeding (proxy)	Tabular baseline	xgboost	0.864	[0.852, 0.875]	0.442	0.338	0.062
		random forest	0.858	[0.846, 0.870]	0.431	0.157	0.064
		mlp	0.764	[0.746, 0.779]	0.267	0.326	0.122
		logistic regression	0.722	[0.703, 0.742]	0.249	0.317	0.114

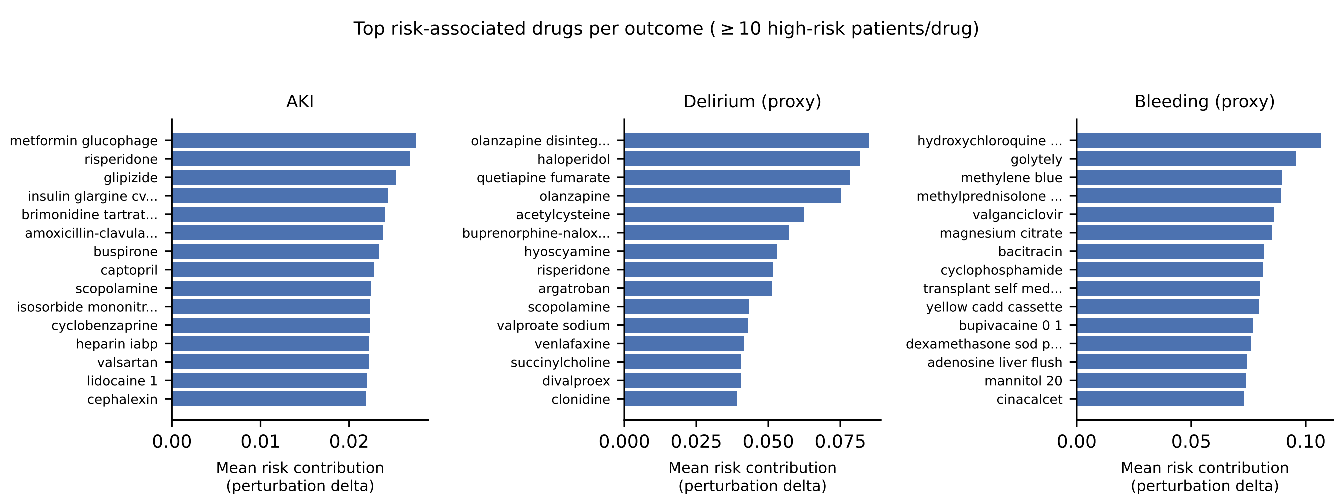

Fig. 4. Top perturbation-ranked risk-associated drugs per outcome (reliability-filtered, ≥10 high-risk patients per drug).

TABLE IV

PAIRED BOOTSTRAP SIGNIFICANCE TEST: NODE2VEC EMBEDDING VS. LEAKAGE-CORRECTED TABULAR BASELINE, DELIRIUM OUTCOME, IDENTICAL 5-FOLD CV SPLITS, 2,000 RESAMPLES.

Quantity	Value
AUROC (embedding)	0.8760
AUROC (tabular, leak-free)	0.8582
Observed ΔAUROC	0.0178
95% CI on Δ	[0.0112, 0.0249]
p (approx., two-sided)	0.0000

TABLE V

CONFIRMATORY REPLICATION ON AN INDEPENDENT, ZERO-OVERLAP SECOND 10,000-ADMISSION SUBSAMPLE (GRAPH AND EMBEDDINGS REBUILT FROM SCRATCH).

Quantity	Original sample	Replication
ΔAUROC (emb. $-$ tab.)	0.0178	0.0152
95% CI	[0.0112, 0.0249]	[0.0081, 0.0218]

with no external validation cohort. (4) No classifier hyperparameter tuning was performed, by design, for a fair fixed-hyperparameter comparison; absolute performance ceilings for any individual model are therefore likely understated. (5) Perturbation-based drug-risk rankings have not been manually cross-checked against a pharmacological drug-drug-interaction reference and should not be read as clinically validated without that step.

VI. CONCLUSION

We compared graph-embedding, graph-theoretic, and tabular feature representations for three polypharmacy-relevant ICU adverse events on MIMIC-IV under a matched protocol. The result is outcome-dependent: tabular features win with unrestricted drug-identity access (AKI, bleeding), while graph embeddings outperform a leakage-corrected tabular baseline for delirium – significant on paired folds and replicated on an independent subsample. We further contribute a documented label-leakage correction and reliability-filtered interpretability outputs. Future work includes node2vec hyperparameter search, external validation, and clinical cross-referencing of the identified drug combinations.

ACKNOWLEDGMENT

The authors thank PhysioNet and the MIMIC-IV database creators for providing credentialed access to the data underlying this work. AI-assisted tools were used for LaTeX typesetting and grammar refinement; all research design, analysis, and conclusions in this paper are solely the authors' own.

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